VS-09 Infrastructure and Platform Enablement

Table of contents

VS-09 Infrastructure and Platform Enablement . . . 1
1. Purpose and value created . . . 1
2. Relation to the value chain . . . 1
3. Stages . . . 2
4. Plan, build, operate, and improve . . . 5
5. Actors and roles . . . 6
6. Capabilities, processes, and functions . . . 7
7. Principles . . . 8
8. Interfaces, services, and events . . . 8
9. Key performance indicators . . . 9
10. Examples . . . 10

VS-09 Infrastructure and Platform Enablement

1. Purpose and value created

VS-09 provides the technical and digital foundation of the statistical organisation. The other value streams work on data, methods, governance, or relationships. VS-09 makes sure that all of this work can actually run: safely, efficiently, and at scale. It delivers the IT environments, data platforms, tools, and integration services that statistical production needs, from data collection to dissemination. It keeps them secure, resilient, and compliant, and flexible enough to adopt new technologies and new data sources as the NSI modernises. The value created is **reliable enablement**, or operational capability and continuity: a stable, modern, and adaptive foundation that turns infrastructure from a background function into an active enabler of the statistical mission.

VS-09 corresponds to the horizontal domain **Infrastructure**. Its signature discipline is that platforms are run as supported, monitored, and recoverable services, with security built in from the start, never as one-off builds. This section describes the business layer of that enablement: the capability of providing platforms and services. The application and technology architecture of the platforms themselves is described in the application and technology layers of the SAF, not here.

PLACEHOLDER: ArchiMate view VS-09 (a) value stream and capabilities

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2. Relation to the value chain

VS-09 is not a stage in the production line. It is the foundation underneath it: every primary and supporting value stream runs on the platforms that VS-09 provides. VS-09 does not take data from one state to the next and has no registration event. It provides the environments in which the states are stored, the gates are run, and the handover events are delivered. It runs continuously, as an always-on service, not per production cycle. Its output is an enabling capability, not a data artefact: platforms, environments, integration services, and assured availability.

VS-09 receives requirements from business capabilities, data volumes, and the other value streams, in particular VS-08 for new technology and VS-06 for dissemination platforms. Its security and continuity posture is governed by VS-07.

Primary value stream(s)	Provides	Receives
VS-01 Data Acquisition and Ingestion	Secure collection and transfer channels, storage, catalogue, event interfaces	Capacity and channel requirements
VS-02 to VS-05 (standardisation, editing, analysis, composition)	Computing and orchestration to process large datasets efficiently; storage and versioning of the states	Capacity demand, processing requirements
VS-06 Dissemination and Service Delivery	Portals, APIs, dashboards, secure research environments	Channel and scaling requirements
All primary value streams	Steady-state event interfaces, assured availability, recovery	Incidents, capacity demand

VS-09 aligns to GAMSO, not to the production phases of GSBPM, and to the supporting business capabilities of the SAF.

GAMSO area or SAF domain	Relevance to VS-09
Capability Management	Manage the IT capability; introduce and modernise technology
Corporate Support	Manage IT services and operations, security, and business continuity
SAF capability domain 3, Supporting business capabilities	3.2 Computerisation and automation (application development, functional management, technical management, IT infrastructure); 3.3 Security and privacy (technical)

PLACEHOLDER: ArchiMate view VS-09 (c) interactions with the value chain: requirements from business capabilities, data volumes, VS-08 (new technology) and VS-06 (channels) flowing into VS-09; environments, platforms, integration, and availability flowing to all value streams VS-01 to VS-10; security and continuity policy from VS-07; incidents and capacity demand flowing back from all streams

3. Stages

The stages form a continuous enablement cycle: define, design, provision, integrate, assure, and evolve. The first two stages plan the foundation, the next two build it, VS-09.5 operates it, and VS-09.6 improves it (see attribute 4).

Stage	Name	Value added
VS-09.1	Define Digital and Technical Requirements	Infrastructure needs follow from business needs
VS-09.2	Design Platform and Infrastructure Architecture	Platforms are designed for storage, processing, integration, and secure access
VS-09.3	Provision and Configure Platforms	Environments are available with the required controls
VS-09.4	Enable Integration and Orchestration Services	Processes and services are connected

Stage	Name	Value added
VS-09.5	Ensure Performance, Security and Continuity	Platforms are reliable and protected
VS-09.6	Evolve and Modernise the Technical Ecosystem	The technical landscape stays current

VS-09.1 Define Digital and Technical Requirements

Value created: *Infrastructure needs follow from business needs.*

The cycle starts from the business, not from technology. The team identifies the infrastructure needs that arise from the business capabilities, the data volumes, and the analytical complexity of the work across the value streams. Requirements come from the consuming value streams: capacity and channel requirements from VS-01, processing demand from VS-02 to VS-05, channel and scaling requirements from VS-06, and new-technology adoption requests from VS-08. The result is a set of infrastructure requirements derived from business needs.

Activity	Key outputs	SAF process	Other reference
Collect requirements from the consuming value streams	Requirements backlog	PRD-2.6 Design production systems and workflows	SAF-P05
Derive infrastructure needs from business capabilities, data volumes, and analytical complexity	Infrastructure requirements	PRD-2.6 Design production systems and workflows	3.2.2 Functional management
Assess requests to adopt new technology from VS-08	Technology onboarding assessment	PRD-2.6 Design production systems and workflows	GAMSO Capability Management

VS-09.2 Design Platform and Infrastructure Architecture

Value created: *Platforms are designed for storage, processing, integration, and secure access.*

The requirements feed the design of the platform and infrastructure architecture. The team produces blueprints for data storage, processing, integration, and secure access, for example a data fabric, a data market, and compute platforms. Reuse of shared and open platform components comes first, in line with SAF-P01. Security and access control are part of the blueprint from the start, in line with SAF-P10.

> **i** On the Medallion architecture
>
> A platform team may use Bronze, Silver, and Gold to organise storage, staging, and table optimisation inside the platform. That is an engineering pattern, not a governance model: it describes how data has been refined, not what the data means, what methodological state it is in, or what another statistical process may rely on. The steady states remain the governance model. Medallion, if used at all, sits underneath it as a storage convention (see *Combining the models*).

Activity	Key outputs	SAF process	Other reference
Design blueprints for storage, processing, integration, and secure access	Architecture blueprints	PRD-2.6 Design production systems and workflows	4.1.7 Architecture Management
Select shared and open platform components before building new ones	Sourcing decision	PRD-2.6 Design production systems and workflows	SAF-P01

Activity	Key outputs	SAF process	Other reference
Design security and access control into the platforms	Security design	PRD-2.6 Design production systems and workflows	SAF-P10; 3.3.2 Security planning
Map platform storage conventions (such as Medallion layers) onto the steady states	Storage convention	PRD-2.6 Design production systems and workflows	

VS-09.3 Provision and Configure Platforms

Value created: *Environments are available with the required controls.*

The architecture is realised by deploying environments, on-premise, hybrid, or in the cloud, with the necessary governance, monitoring, and security controls. This covers the production environments of the primary value streams, the portals, APIs, and dashboards of VS-06, the secure research environments for microdata access, and the sandbox and production runtime that VS-08 needs for new technology. Platforms are self-service and well documented, so that technology serves all teams (SAF-P05).

Activity	Key outputs	SAF process	Other reference
Deploy environments (on-premise, hybrid, cloud)	Deployed environments	PRD-3.4 Configure workflows	3.2.5 IT infrastructure
Configure governance, monitoring, and security controls	Controlled environments	PRD-3.4 Configure workflows	SAF-P10; 3.3.4 Logical identity and access control
Provide secure research environments for microdata access	Secure processing environment	MDS-4 Provide a secure processing environment	
Provide sandbox and production runtime for VS-08	Research and production environments	PRD-3.4 Configure workflows	
Document platforms for self-service use	Platform documentation	PRD-3.4 Configure workflows	SAF-P05

VS-09.4 Enable Integration and Orchestration Services

Value created: *Processes and services are connected.*

This stage provides the APIs, workflow engines, and automation that connect the business services and processes. It includes the steady-state event interfaces that the primary value streams rely on: the registration events, such as *Raw Data Registered*, that trigger the next value stream. It also includes the connectivity to data spaces.

Activity	Key outputs	SAF process	Other reference
Provide APIs and workflow engines	Integration and orchestration services	PRD-3.4 Configure workflows	3.2.1 Application development
Provide the steady-state event interfaces	Event interfaces	PRD-3.4 Configure workflows	SAF-P07
Operate data space connectivity	Data space connectors	DSP-4 Exchange data under usage control	

VS-09.5 Ensure Performance, Security and Continuity

Value created: *Platforms are reliable and protected.*

Day to day, VS-09 monitors platform performance, implements data protection, manages incidents, and guarantees availability and disaster recovery. This is the assurance that the whole chain can run reliably (SAF-P06). Platforms are run as actively supported, monitored services (SAF-P02). Security controls and continuity are assessed with VS-07, which sets the security and continuity policy.

Activity	Key outputs	SAF process	Other reference
Monitor platform performance and capacity	Performance monitoring	PRD-3.5 Test production systems	SAF-P02; 3.2.3 Technical management
Implement data protection: backup and recovery of stored versions	Backups, recovery procedures	DLM-5 Store and version data	3.3.6 IT security services
Manage incidents and restore service	Incident records	PRD-3.5 Test production systems	3.3.3 Security incident management
Guarantee availability and disaster recovery	Availability and disaster recovery plans	DLM-5 Store and version data	SAF-P06
Assess the security posture with VS-07	Security and continuity evidence	PRD-3.5 Test production systems	SAF-P10

VS-09.6 Evolve and Modernise the Technical Ecosystem

Value created: *The technical landscape stays current.*

The cycle ends by evolving the technical ecosystem. New technologies are introduced, often transferred from VS-08. Obsolete components are retired. The foundation stays flexible for new data sources and operating models. The result is a modernisation roadmap and adopted technologies, which feed the next round of requirements.

Activity	Key outputs	SAF process	Other reference
Introduce new technologies, often transferred from VS-08	Adopted technologies	No reference process (NSI-specific)	FUN-2.2 Develop capability improvements
Retire obsolete components	Retirement records	No reference process (NSI-specific)	SAF-P03
Maintain the modernisation roadmap	Modernisation roadmap	No reference process (NSI-specific)	4.1.3 Change management

4. Plan, build, operate, and improve

VS-09 is itself a capability that is designed, built, operated, and improved. Unlike a primary value stream it produces platforms and assured availability, not statistical data, and it runs as an always-on service rather than a per-cycle production run. Its security and continuity are governed by VS-07.

Cycle step	What happens in VS-09	Main outputs	SAF process
Plan	Define requirements and design the architecture and operating model (VS-09.1, VS-09.2)	Infrastructure requirements, architecture blueprints	PRD-2.6 Design production systems and workflows

Cycle step	What happens in VS-09	Main outputs	SAF process
Build	Provision platforms and enable integration (VS-09.3, VS-09.4)	Deployed environments, integration services, event interfaces	PRD-3.4 Configure workflows; MDS-4 Provide a secure processing environment; DSP-4 Exchange data under usage control
Operate	Continuous monitoring, security, and incident management (VS-09.5)	Monitoring, security, availability, disaster recovery	PRD-3.5 Test production systems; DLM-5 Store and version data
Improve	Evolve and modernise the technical ecosystem (VS-09.6)	Modernisation roadmap, adopted technologies	No reference process (NSI-specific)

VS-09 manages enabling artefacts: technical requirements, architecture blueprints, provisioned environments and platforms, integration and orchestration services and event interfaces, monitoring, security, and continuity controls, and the modernisation roadmap. These artefacts underpin every value stream and are managed across their life cycle (SAF-P03), with security (SAF-P10) and continuity (SAF-P06) built in. Each artefact has a designated owner. Changes are tracked, prior versions remain identifiable, and obsolete components are formally retired.

5. Actors and roles

Specialised role names used in VS-09 map onto the reference roles (see *Role names in the value streams*). The platform engineer role is anchored in VS-09.

Actor or role	ID	Process type	Responsibility in VS-09
Chief technology or infrastructure officer	ACT-01; member of COL-07	Management	Accountable for the technical foundation and its strategy
Supplier and service provider	ACT-09	External	Cloud and hosting providers and software vendors that deliver platform services under contracts with security, confidentiality, and continuity requirements
Process owners of VS-01 to VS-10	ROL-01	Management	Define requirements as consuming teams and run their processes on the platforms
Data custodian	ROL-04	Execution	Runs storage, backup, and access control according to the decisions of the data owner
Data engineer (integration engineer)	ROL-10	Implementation	Builds APIs, workflow engines, and orchestration
Platform engineer (platform or cloud architect)	ROL-11	Design	Designs the platform and infrastructure architecture
Platform engineer (DevOps or platform engineer)	ROL-11	Implementation	Provisions, configures, and operates environments

Actor or role	ID	Process type	Responsibility in VS-09
Platform engineer (service desk or operations)	ROL-11	Execution	Runs incident management and keeps services available
Security officer (security and continuity officer)	ROL-20	Management	Implements security controls, monitoring, and disaster recovery, with VS-07
Enterprise architect	ROL-24	Design, Management	Reviews the platform architecture against the architecture principles in COL-07 Architecture board

6. Capabilities, processes, and functions

Capability foundations

Capability group	Capabilities	Role in VS-09
2.1 Data management	2.1.1 Data lifecycle management	Store, version, back up, and recover the states
3.2 Computerisation and automation	3.2.1 Application development, 3.2.2 Functional management, 3.2.3 Technical management, 3.2.5 IT infrastructure	Develop, configure, and manage platforms and integration services
3.3 Security and privacy	3.3.2 Security planning, 3.3.3 Security incident management, 3.3.4 Logical identity and access control, 3.3.6 IT security services	Build technical security into every platform and respond to incidents
4.1 Strategy and leadership	4.1.3 Change management, 4.1.7 Architecture Management	Govern the platform architecture and its modernisation

Stages, processes, and functions

Stage	Realised by processes	Main functions
VS-09.1 Define Digital and Technical Requirements	PRD-2.6 Design production systems and workflows	FUN-2.5 Manage enterprise architecture; FUN-3.1 Design and build statistical products
VS-09.2 Design Platform and Infrastructure Architecture	PRD-2.6 Design production systems and workflows	FUN-2.5 Manage enterprise architecture; FUN-10.3 Manage information technology
VS-09.3 Provision and Configure Platforms	PRD-3.4 Configure workflows; MDS-4 Provide a secure processing environment	FUN-8.3 Operate secure research environments; FUN-10.3 Manage information technology
VS-09.4 Enable Integration and Orchestration Services	PRD-3.4 Configure workflows; DSP-4 Exchange data under usage control	FUN-9.2 Operate data space connectivity; FUN-10.3 Manage information technology
VS-09.5 Ensure Performance, Security and Continuity	PRD-3.5 Test production systems; DLM-5 Store and version data	FUN-10.3 Manage information technology; FUN-10.8 Manage information security and privacy
VS-09.6 Evolve and Modernise the Technical Ecosystem	No reference process (NSI-specific)	FUN-2.2 Develop capability improvements; FUN-10.3 Manage information technology

PLACEHOLDER: ArchiMate view VS-09 (b) stages realised by processes and functions

Framework mappings

Cycle step	SAF business processes	GSBPM	GAMSO	SAF capabilities
Plan	PRD-2.6	2.6 Design production systems and workflow	Capability Management	4.1.7 Architecture Management, 3.2.2 Functional management
Build	PRD-3.4, MDS-4, DSP-4	3.4 Configure workflows	Capability Management	3.2.1 Application development, 3.2.5 IT infrastructure
Operate	PRD-3.5, DLM-5	3.5 Test production system	Corporate Support (IT services, security, continuity)	3.2.3 Technical management, 3.3.3 Security incident management, 3.3.6 IT security services
Improve	None (NSI-specific)		Capability Management	4.1.3 Change management

7. Principles

VS-09 foregrounds SAF-P10, SAF-P06, SAF-P02, and SAF-P05.

Principle	Statement	Application in VS-09
SAF-P10	Security By Design	Security and access control are built into every platform and environment from the start.
SAF-P06	Business Continuity	VS-09 guarantees availability, resilience, and disaster recovery for the whole chain.
SAF-P02	Active support	Platforms are run as actively supported, monitored services, not one-off builds.
SAF-P05	IT and technology is everyone's business	Platforms are self-service and well documented, so that technology serves all teams.
SAF-P01	Actively participate in open-source communities to maintain collective control over statistical processes	Open, shared platform components are preferred, so that the NSI keeps collective control.
SAF-P03	Active Life Cycle management	Enabling artefacts have an owner and a version history; obsolete components are formally retired.

8. Interfaces, services, and events

Business services

Service	Offered to	Realised in	Interfaces
Platform and environment provisioning service	All value streams VS-01 to VS-10	VS-09.3 Provision and Configure Platforms	Platform or service request
Integration and orchestration service	All primary value streams	VS-09.4 Enable Integration and Orchestration Services	APIs, workflow engines, steady-state event interfaces, data space connectors
Dissemination platform service	VS-06 Dissemination and Service Delivery	VS-09.3 Provision and Configure Platforms	Portals, APIs, dashboards, secure research environments
Research and innovation runtime service	VS-08 Innovation and Methodological Development	VS-09.3 Provision and Configure Platforms	Sandbox and production runtime for new technology
Platform operations and continuity service	All value streams	VS-09.5 Ensure Performance, Security and Continuity	Service desk, incident management, monitoring

Business events

Event	Triggers or marks
Platform or service request	A consuming value stream asks for an environment or service; input to VS-09.1
Capacity change	A change in capacity demand; review of capacity in VS-09.1 and VS-09.5
Incident	Start of incident management in VS-09.5
Dissemination platform request	VS-06 asks for a new or scaled channel
Technology onboarding	VS-08 asks to adopt a new technology; input to VS-09.1 and VS-09.6
Security or continuity review	Periodic or triggered review of the security and continuity posture with VS-07

Interaction with the primary value streams

VS-09 interacts with all primary value streams at once (see attribute 2). The interaction runs through *Platform or service request*, *Incident*, and *Capacity change*; VS-06 also uses *Dissemination platform request*.

Interaction with the supporting value streams

Value stream	Provides to VS-09	Receives from VS-09	Event
VS-07 Governance and Compliance	Security and continuity policy to enforce	Security posture, continuity and incident evidence	*Security or continuity review*
VS-08 Innovation and Methodological Development	New-technology adoption requests; requirements for sandbox and production platforms	Sandbox and production runtime for new technology	*Technology onboarding*; *Platform or service request*

9. Key performance indicators

KPI	Description	Measurement
Platform availability	Measured availability against the agreed service level, per platform	Monthly
Incident response	Mean time to restore service after an incident, per severity class	Per incident, trended quarterly
Event-interface reliability	Delivery rate of the steady-state handover events the chain depends on	Per interface, monthly
Capacity headroom	Share of platforms operating within their planned capacity envelope	Monthly, ahead of known peaks
Security posture	Open critical vulnerabilities and patch latency, assessed with VS-07	Continuous, reported quarterly
Modernisation progress	Components retired or replaced according to the roadmap	Annually

10. Examples

PLACEHOLDER: VS-09 examples to be added, for example provisioning a secure research environment, an outage of the event interface and its recovery, and the transfer of a new technology from VS-08 into production

Toolkit: SAF-TK-V09 VS-09 maturity checklist and scenarios